Lecture 31

Section 6.2

Orthogonal Sets

Definition: A set of vectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ in $\mathbb{R}^n$ is said to be an orthogonal set if $\mathbf{u}_i \bullet \mathbf{u}_j = 0$ for all $i \neq j$.

Example: Let

$$\mathbf{u}_1 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_2 = \begin{pmatrix} 2 \\ -4 \\ -2 \end{pmatrix}, \qquad \mathbf{u}_3 = \begin{pmatrix} -2 \\ -8 \\ 14 \end{pmatrix}.$$

Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ form an orthogonal set.

Solution: We see that

$$\begin{aligned}
\mathbf{u}_1 \bullet \mathbf{u}_2 &= 3 \cdot 2 + 1 \cdot (-4) + 1 \cdot (-2) = 6 - 4 - 2 = 0, \\
\mathbf{u}_1 \bullet \mathbf{u}_3 &= 3 \cdot (-2) + 1 \cdot (-8) + 1 \cdot 14 = -6 - 8 + 14 = 0, \\
\mathbf{u}_2 \bullet \mathbf{u}_3 &= 2 \cdot (-2) + (-4) \cdot (-8) + (-2) \cdot 14 = -4 + 32 - 28 = 0.
\end{aligned}$$

Theorem: If $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent. Therefore, S is a basis for the subspace spanned by S.

Proof: Let's suppose that

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_m\mathbf{u}_m = \mathbf{0},$$

where $c_1, \ldots, c_m$ are scalars. Let's dot this with $\mathbf{u}_1$. We have that

$$\begin{aligned}
0 &= \mathbf{0} \bullet \mathbf{u}_1 \\
&= (c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_m\mathbf{u}_m) \bullet \mathbf{u}_1 \\
&= c_1\mathbf{u}_1 \bullet \mathbf{u}_1 + c_2\mathbf{u}_2 \bullet \mathbf{u}_1 + \cdots + c_m\mathbf{u}_m \bullet \mathbf{u}_1 \\
&= c_1(\mathbf{u}_1 \bullet \mathbf{u}_1) + c_2(\mathbf{u}_2 \bullet \mathbf{u}_1) + \cdots + c_m(\mathbf{u}_m \bullet \mathbf{u}_1).
\end{aligned}$$

Because S is orthogonal, we have that $\mathbf{u}_i \bullet \mathbf{u}_j = 0$ for all $i \neq j$. So,

$$c_1(\mathbf{u}_1 \bullet \mathbf{u}_1) + c_2(\mathbf{u}_2 \bullet \mathbf{u}_1) + \cdots + c_m(\mathbf{u}_m \bullet \mathbf{u}_1) = c_1(\mathbf{u}_1 \bullet \mathbf{u}_1).$$

Because $\mathbf{u}_1$ is nonzero, then $\mathbf{u}_1 \bullet \mathbf{u}_1 \neq 0$. Therefore, since $0 = c_1(\mathbf{u}_1 \bullet \mathbf{u}_1)$, it follows that $c_1 = 0$. We repeat this computation $m - 1$ more times by dotting with $\mathbf{u}_2, \ldots, \mathbf{u}_m$ instead, showing that $c_2, \ldots, c_m$ are all equal to 0.

Definition: An orthogonal basis for a subspace W of $\mathbb{R}^n$ is a basis for W that is an orthogonal set.

Example: The standard basis is an orthogonal basis for $\mathbb{R}^n$.

Theorem: Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y} \in W$, there is a unique linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_m\mathbf{u}_m$$

where

$$c_j = \frac{\mathbf{y} \bullet \mathbf{u}_j}{\mathbf{u}_j \bullet \mathbf{u}_j}$$

for $j = 1, \ldots, m$.

Proof: Use the same idea as in the previous (theorem) proof.

Example: Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, as in the previous example. We showed that this set is orthogonal. Hence, it is a basis (for $\mathbb{R}^3$). Let's express

$$\mathbf{y} = \begin{pmatrix} -2 \\ -18 \\ 46 \end{pmatrix}$$

as a linear combination of vectors in S.

Solution: We see that

$$\begin{aligned}
c_1 &= \frac{\mathbf{y} \bullet \mathbf{u}_1}{\mathbf{u}_1 \bullet \mathbf{u}_1} = \frac{(-2)\cdot 3 + (-18)\cdot 1 + 46 \cdot 1}{3 \cdot 3 + 1 \cdot 1 + 1 \cdot 1} = \frac{-6 - 18 + 46}{9 + 1 + 1} = \frac{22}{11} = 2, \\
c_2 &= \frac{\mathbf{y} \bullet \mathbf{u}_2}{\mathbf{u}_2 \bullet \mathbf{u}_2} = \frac{(-2)\cdot 2 + (-18)\cdot(-4) + 46 \cdot (-2)}{2 \cdot 2 + (-4)\cdot(-4) + (-2)\cdot(-2)} = \frac{-4 + 72 - 92}{4 + 16 + 4} = -\frac{24}{24} = -1, \\
c_3 &= \frac{\mathbf{y} \bullet \mathbf{u}_3}{\mathbf{u}_3 \bullet \mathbf{u}_3} = \frac{(-2)\cdot(-2) + (-18)\cdot(-8) + 46 \cdot 14}{(-2)^2 + (-8)^2 + 14^2} = \frac{4 + 144 + 644}{4 + 64 + 196} = \frac{792}{264} = 3.
\end{aligned}$$

So,

$$\mathbf{y} = 2\mathbf{u}_1 - \mathbf{u}_2 + 3\mathbf{u}_3.$$

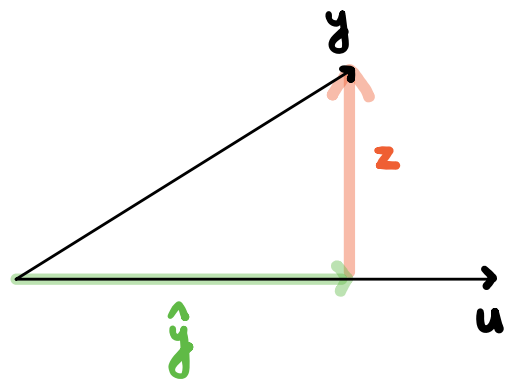
Note: Observe how nice it is to compute the weights of $\mathbf{y}$ from an orthogonal basis.

Orthogonal Projection

Suppose $\mathbf{u}$ is a nonzero vector of $\mathbb{R}^n$. Given any other vector $\mathbf{y}$ in $\mathbb{R}^n$, we want to decompose $\mathbf{y}$ as

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}} = \alpha\mathbf{u}$ for some scalar α and $\mathbf{z}$ is a vector orthogonal to $\mathbf{u}$.



Definition: The vector $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$. The vector $\mathbf{z}$ is the component of $\mathbf{y}$ orthogonal to $\mathbf{u}$.

If c is a nonzero scalar, then the orthogonal projection of $\mathbf{y}$ onto $c\mathbf{u}$ is exactly the same as the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$. So, $\hat{\mathbf{y}}$ is determined by $L = \text{Span}\{\mathbf{u}\}$, i.e. the line through $\mathbf{u}$ and $\mathbf{0}$. So, we can write $\hat{\mathbf{y}} = \text{proj}_L\mathbf{y}$, namely the orthogonal projection of $\mathbf{y}$ onto L.

Question: How do we find $\hat{\mathbf{y}}$ and $\mathbf{z}$?

From before, we know that $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$, i.e. $\mathbf{z} = \mathbf{y} - \alpha\mathbf{u}$. Then, $\mathbf{z}$ is orthogonal to $\mathbf{u}$ if and only if

$$0 = \mathbf{z} \bullet \mathbf{u} = (\mathbf{y} - \alpha\mathbf{u}) \bullet \mathbf{u} = \mathbf{y} \bullet \mathbf{u} - \alpha(\mathbf{u} \bullet \mathbf{u}).$$

So, in order for $\mathbf{u}$ and $\mathbf{z}$ to be orthogonal, we must have that

$$\alpha = \frac{\mathbf{y} \bullet \mathbf{u}}{\mathbf{u} \bullet \mathbf{u}}.$$

Putting this all together,

$$\hat{\mathbf{y}} = \text{proj}_L\mathbf{y} = \left(\frac{\mathbf{y} \bullet \mathbf{u}}{\mathbf{u} \bullet \mathbf{u}}\right)\mathbf{u}.$$

Note that the point identified with $\hat{\mathbf{y}}$ is the closest point of L to $\mathbf{y}$.

Example: Let

$$\mathbf{y} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \qquad \text{and} \qquad \mathbf{u} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$

Find the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$. Also, find the distance from $\mathbf{y}$ to L, where L is the line passing through $\mathbf{u}$.

Solution: We see that

$$\hat{\mathbf{y}} = \left(\frac{\mathbf{y} \bullet \mathbf{u}}{\mathbf{u} \bullet \mathbf{u}}\right)\mathbf{u} = \left(\frac{4 \cdot 2 + 1 \cdot 5}{2 \cdot 2 + 5 \cdot 5}\right)\begin{bmatrix} 2 \\ 5 \end{bmatrix} = \frac{13}{29} \cdot \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 26/29 \\ 65/29 \end{bmatrix}.$$

To find the distance from $\mathbf{y}$ to L, we must first find $\mathbf{y} - \hat{\mathbf{y}} = \mathbf{z}$. We see that

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} - \begin{bmatrix} 26/29 \\ 65/29 \end{bmatrix} = \begin{bmatrix} 90/29 \\ -36/29 \end{bmatrix}.$$

So, to compute the distance from $\mathbf{y}$ to L, we need to compute $||\mathbf{z}||$. Therefore,

$$||\mathbf{z}|| = \sqrt{\mathbf{z} \bullet \mathbf{z}} = \sqrt{\left(\frac{90}{29}\right)^2 + \left(-\frac{36}{29}\right)^2} = \frac{1}{29}\sqrt{90^2 + (-36)^2} = \frac{1}{29}\sqrt{9396}.$$

As an aside, notice that we now have the information needed to write $\mathbf{y}$ as a sum of $\hat{\mathbf{y}}$ and $\mathbf{z}$.

Orthonormal Sets

Definition: A set $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ is an orthonormal set if it is an orthogonal set of unit vectors.

Definition: An orthonormal basis of a subspace W of $\mathbb{R}^n$ is a basis (of W) which is also an orthonormal set.

Note: We can obtain an orthonormal set from an orthogonal set by normalizing the vectors in the orthogonal set.

Theorem: An $m \times n$ matrix A has orthonormal columns if and only if $A^T A = I$.

Theorem: Let A be an $m \times n$ matrix with orthonormal columns. Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then,

1. $||A\mathbf{x}|| = ||\mathbf{x}||$,
2. $(A\mathbf{x}) \bullet (A\mathbf{y}) = \mathbf{x} \bullet \mathbf{y}$,
3. $(A\mathbf{x}) \bullet (A\mathbf{y}) = 0$ if and only if $\mathbf{x} \bullet \mathbf{y} = 0$.

In other words, $\mathbf{x} \mapsto A\mathbf{x}$ preserves lengths and orthogonality.

Note: The two prior theorems are especially useful when $m = n$. An orthogonal matrix is a square invertible matrix A such that $A^{-1} = A^T$. Note that A is orthogonal if and only if A has orthonormal columns.